

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM

233/3



CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- Answer all the questions in this paper in the spaces provided.
- You are not allowed to start working with the apparatus for the first 15 minutes of the $2\frac{1}{4}$ hours allowed for this paper.
- This time is to enable you to read the question paper and make sure you have all the apparatus and chemicals that you may need.
- All working **must** be clearly shown where necessary.
- Mathematical tables/electronic calculators may be used.

FOR EXAMINERS USE ONLY

Question	Maximum score	Candidate's score
1	21	
2	08	
3	11	
Total score	40	

1.

You are provided with;

-Solution A, containing 4.0 gdm^{-3} of sodium hydroxide

- Solution B, hydrochloric acid

- 2.5g of a mixture of two salts, XCl (RFM=58.5) and X_2CO_3 (RFM=106)

You are required to:

i) Standardize solution B, hydrochloric acid.

ii) Determine the percentage composition of XCl salt in the mixture.

PROCEDURE I

1. Fill the burette with solution B

2. Pipette solution A into a clean dry conical flask. Then add 2 -3 drops of phenolphthalein indicator.

3. Titrate solution B against solution A. Record your results in the table below.

4. Repeat the procedure two more times and complete the table 1.

TABLE I

Titration number	1	2	3
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of acid used(cm^3)			

(4marks)

a) Calculate the average volume of solution B used.

(1 mark)

b) Calculate;

i) Moles of sodium hydroxide that reacted with the acid. (Na=23.0, O= 16.0, H=1.0) (2marks)

ii) Moles of hydrochloric acid in the average volume.

(1 mark)

iii) Molarity of the acid

(1 mark)

PROCEDURE II

1. Put about 100cm³ of water in a 250ml volumetric flask add all solid C, the 2.5g of salt mixture. Shake the mixture to dissolve the solid. Top up the solution to the mark with distilled water, label this solution C.
2. Fill this burette with solution B.
3. Pipette 25cm³ of solution C and put it into a clean conical flask. Add 3 drops of methyl orange indicator.
4. Titrate solution C with solution B. Record your results in the table below.
5. Repeat the titration two more times and complete the table II.

TABLE II

TITRATION	1	2	3
Final burette reading(cm ³)			
Initial burette reading (cm ³)			
Volume of the solution B used (cm ³)			

(4marks)

c). Calculate the average volume of solution **B**

(1 mark)

d). Calculate the number of moles in hydrochloric acid used.

(1 mark)

e) The equation for the reaction of the acid with one of the salts in the mixture is;



Calculate;

i). Moles of X_2CO_3 that reacted with the acid in the experiment.

(1

mark)

ii) Molarity of X_2CO_3

(1marks)

f) Calculate the mass of the salt in gdm^{-3}

(1 mark)

g) Calculate the percentage of XCl in the mixture.

(3

marks)

2. In this experiment you're required to determine the time taken for a precipitate to be formed when S_3 which is sodium thiosulphate solution, reacts with dilute hydrochloric acid.

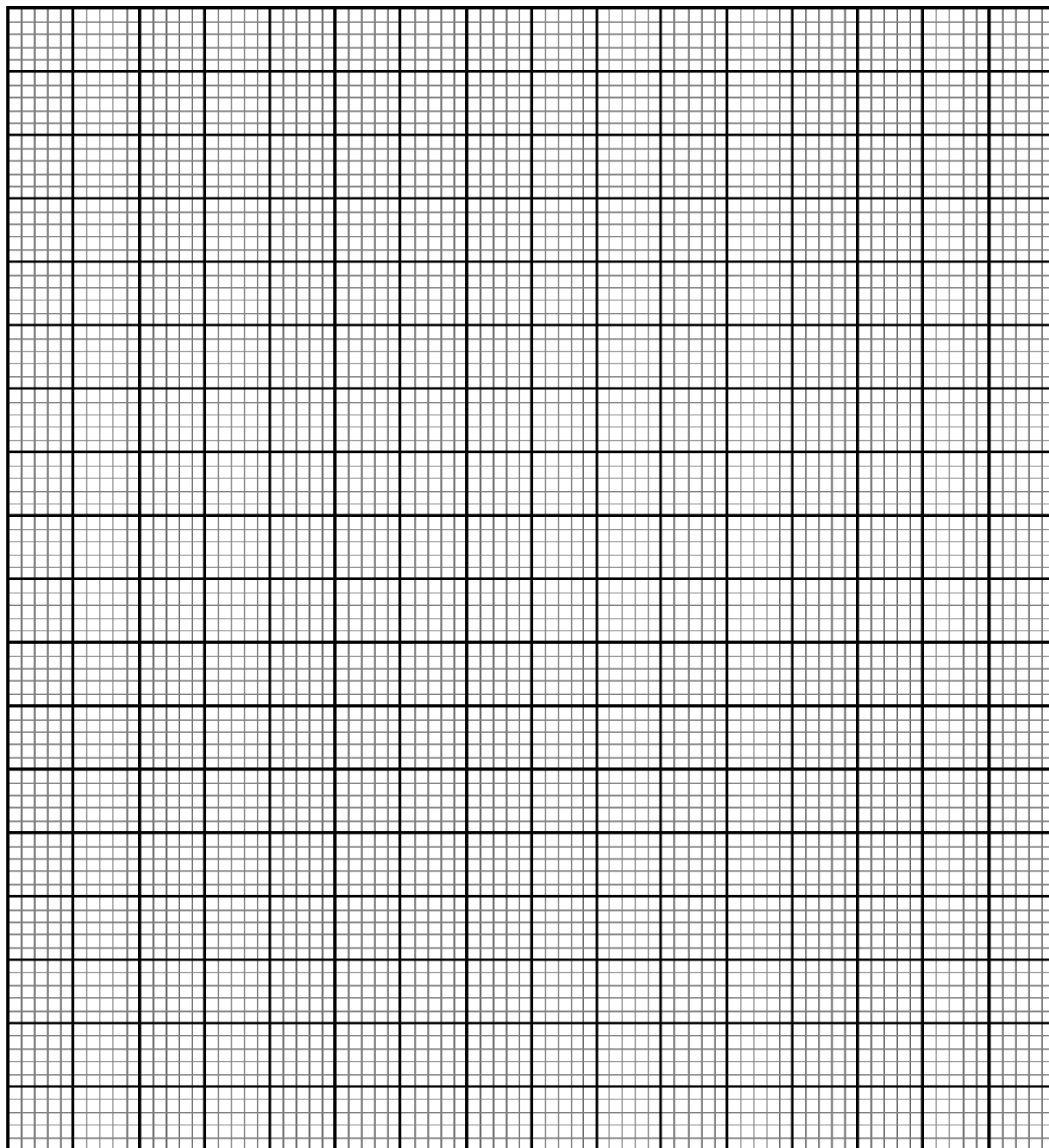
PROCEDURE

1. Using a measuring cylinder measure 50cm^3 of S_3 into a 100ml beaker.
2. Make a pencil cross on the white piece of paper so that a beaker is placed on top of the paper, the cross can be seen through the bottom of the beaker.
3. To solution A add 10cm^3 of 2M hydrochloric acid using 10ml measuring cylinder and at the same time start a stop watch/clock. Swirl the contents of the beaker twice and then place it over the cross on the paper. Look at the cross from above the beaker through the mixture. Stop the watch immediately the precipitate makes the cross invisible. Record the time taken for the cross to become invisible in the table below, rinse the beaker.
4. Repeat the procedure with solutions B, C, D and E as per the table.

SOLUTION	Volume of solution S_3 in the beaker(cm^3)	Volume of water added(cm^3)	Volume of 2M HCl	Time taken in seconds
A	50	0	10	
B	40	10	10	
C	30	20	10	
D	20	30	10	
E	10	40	10	

(3 marks)

- a) Plot the graph of volume of solution S_3 (y-axis) against time. (3 marks)



b) From the graph state the relationship between concentration of the solution S_3 and time **(1mk)**

.....
.....

c) Calculate the reaction rate in solution C in the table above.

(1mk)

.....

.....

.....

a) Heat a spatula end full of solid D in a clean dry test-tube.

Observation	Inference
(1mk)	(½mk)

b) Put the remaining solid D in a boiling tube. Add about 10cm³ distilled water. Shake the mixture.

Observation	Inference
($\frac{1}{2}$ mk)	(1mk)

c). Divide the resultant mixture in (b) above into 5 portions.

i) In the first portion add 2 cm³ of lead (ii) nitrate and then warm the mixture

Observation	Inference
(1mk)	(1mk)

ii) To the 2nd portion and dilute nitric acid followed by a few drops of Barium nitrate.

Observation	Inference
(1/2mk)	(1/2mk)

iii) To the third portion, add sodium hydroxide solution dropwise until in excess.

Observation	Inference

i) To about 1cm³ of liquid E in a test-tube, add about 1cm³ of distilled water and shake the mixture.

ii). To about 1cm³ of liquid E in a test-tube add a small amount of solid sodium hydrogen carbonate.

iii). To about 2cm³ of liquid E in a test-tube, add about 1cm³ of acidified potassium dichromate(vi). Warm the mixture gently and allow it to stand for about one minute.

For Marking Schemes Contact 0746 222 000 / 0742 999 000